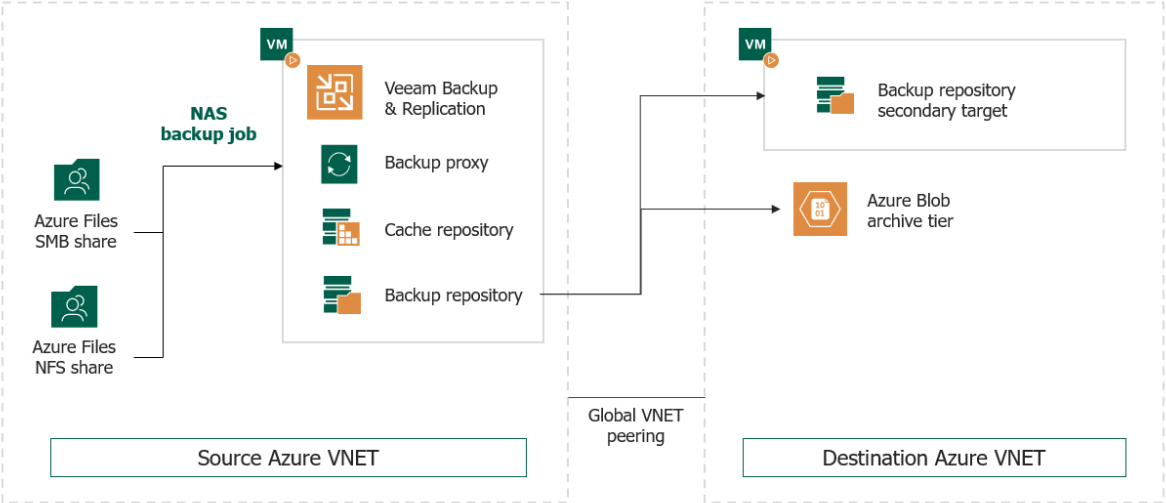
Solution components



Implementation Steps

1. **Deploy Veeam Backup and Replication in Azure.**

   Considerations:

   a. If the VBR installed on a Linux Platform in Azure, a separate Proxy Azure Instance should be created permanently. Required sizing 4vCPU/8GB.
   b. If the VBR installed on a Windows Platform – Proxy can be within the same Azure VBR instance
   c. If you must use Veeam Backup & Replication on premises to protect an Azure File share, place the proxy and primary repository in Azure, and ensure the appropriate connectivity exists in and out of Azure.

1. **Configure Backup Repository**

   1. Backup repository should be in Azure Cloud.
   2. Veeam Data vault which is created on Azure can be used as backup repository.

   Determine if you wish to define a secondary target or store data for archival purposes.

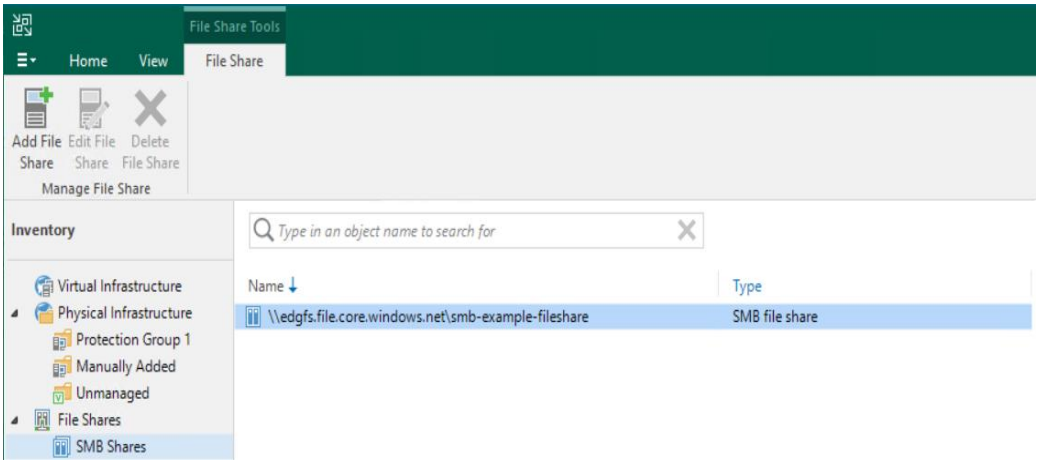
   3. A secondary target can be set on jobs to make duplicate copy on a different repository in a different location.
   4. An archive repository on Azure Blob can be used to store data no longer covered by the primary retention policy.

5. You can consider Scale -out backup repository

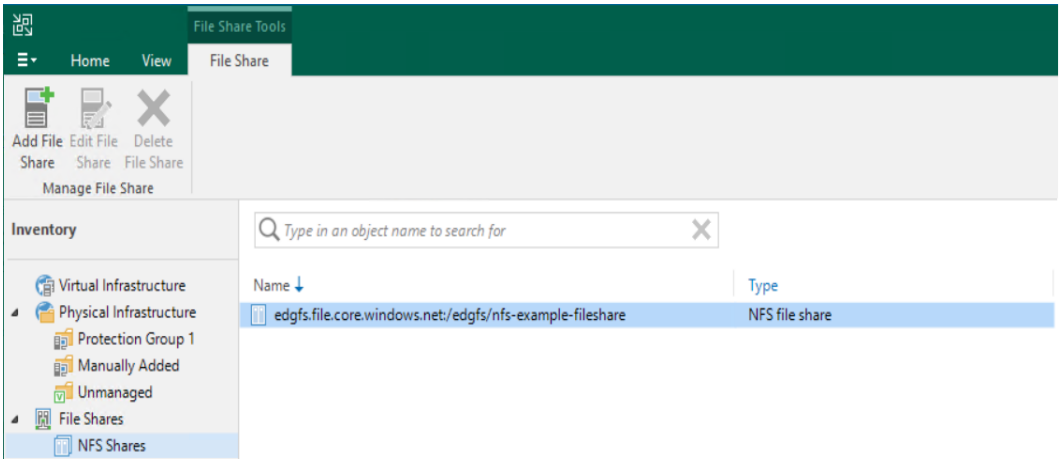
- Performance tier: fast disks on Azure VM disks
- Capacity tier: Azure Blob for long retention / cost optimization

2. Add the Azure file share under Veeam Backup and Replication Inventory.

a. For the SMB share option, you will be asked to provide the UNC path of your file share and specify access credentials. Under advanced options, select 'backup directly from the file share'. Note: Azure Files does not support VSS integration.



b. For the NFS share option, you will need to provide the path in server:/folder format. In Azure, NFS file shares can only be accessed from trusted networks, so a virtual network rule must be configured. As NFS is considered less secure you may have to change the storage account configuration to not require secure transfer.



3. **Backup**
   - ➢ One Full backup.
   - ➢ Forever Incremental – CFT mechanism will be used for Incremental backup

CFT most commonly refers to Changed File Tracking — the mechanism Veeam uses for file share (NAS) backups to detect what files changed since the last run, so incremental don't require re-reading everything.

- Keeps a "state" of what was backed up previously (file metadata and identifiers).
- On the next run, it determines **new/changed/deleted** files and backs up only what's needed.
- This is critical for performance, especially with **millions of small files**.
- Subsequent runs rely on CFT and are usually faster—unless the dataset has very high churn (common with Kubernetes workloads writing many small files).

4. **Restore**
   If your AKS/Kubernetes workloads write data to an Azure File Share (typically via CSI) and you protect that share with Veeam Backup & Replication File Share backup, then VBR supports file-level restores, not Kubernetes-aware restores
   - Individual files
   - Folders/directories
   - Entire share contents (practically: restore a large folder tree)

   Restore to:
   - Original location (same Azure File Share path)
   - Alternate location (another share or a different path)

   Key Notes:
   - Veeam uses Azure snapshot technology for fast and efficient backups.
   - You can restore individual files or entire shares.
   - For cross-tenant restores (restoring to a different Azure tenant), you may need additional permissions and setup.
5. **License**
   500GB of NAS/File share capacity = 1 VUL license
   Please refer the below link for license info
   https://www.veeam.com/products/buy/universal-license.html

Step-by-step Procedure: -

- **Adding the file share to VBR: -**
  1. Open **Veeam Backup & Replication Console**
  2. Go to **Inventory**
  3. Select **File Shares** (or **NAS and File Shares**, depending on version)
  4. Click **Add** → choose **SMB share**
  5. Enter the share path in UNC format:
     \\<storageaccount>.file.core.windows.net\<sharename>
  6. Click **Add...** to provide credentials:
     - If using storage account key auth:
       - **Username: AZURE\<storageaccountname>**
       - **Password:** the **storage account access key**
     - If using AD-integrated auth:
       - Use the appropriate **domain user** that has share permissions
  7. Finish the wizard and let VBR **scan/index** the share
     After this, you can create a **File Share Backup Job** and select the added share.

- **Create the backup job and add the share**
  1. Open **Veeam Backup & Replication Console**
  2. Go to **Home**
  3. Click **Backup Job** → **File share...** (wording may show as **File Share** / **NAS Backup** depending on VBR version)
  4. In **Job name**, enter a name (e.g., **AFS-<sharename>-Daily**)
  5. On the **Storage** step:
     - Choose the **Backup repository** (ideally fast disk / hardened repo if using immutability)
     - Configure **Retention** (how many restore points/days to keep)
     - (Optional) enable **Backup file encryption** if required by policy
  6. On the **Schedule** step:
     - Enable schedule (daily/weekly/etc.)
     - Set the backup window appropriate for your environment
  7. Finish the wizard
  8. Right-click the job → **Start** to run it once and validate

- **Restore steps for an Azure File Share protected by VBR File Share (NAS) backup**

1) Start the restore wizard

1. Open **Veeam Backup & Replication Console**
2. Go to **Home**
3. Click **Restore** (top ribbon)
4. Choose **File share** (sometimes shown as **NAS** or **File share** restore depending on VBR version)

2) Select what to restore

5. Pick the backup source/job (your Azure File Share backup)
6. Select the **restore point** (date/time)
7. In the browser, select:
    - **Files** for single file restore, or
    - **Folders** for folder/tree restore

3) Choose restore mode

You'll typically see one of these options:

Option A — Restore to original location

8. Select **Restore to original location**
9. Choose restore behaviour if prompted:
    - **Keep existing files** / **Overwrite** / **Rename**
10. Finish → **Restore**

Use this when you want data back in the same Azure File Share path.

Option B — Restore to an alternate location (recommended for Kubernetes data)

8. Select **Restore to a new location** / **Restore to alternate path**
9. Specify the target path, for example:
    - Another Azure File Share: **\\<storageaccount>.file.core.windows.net\<othershare>\restore_staging**
    - Or a different folder on the same share
10. Provide credentials for the target share/path if asked

11. Finish → **Restore**

This is safer when AKS pods might be actively writing to the original location.

4) Monitor and validate

12. Go to **History → Sessions** (or just watch the running session popup)
13. Verify the restored content:

- Check file counts/sizes
- Validate permissions/ACLs if your apps rely on them
- For AKS: scale workloads back up and confirm read/write access